

Step 1

Setup an OCI cloud account and an instance of a transactional database. See instructions here: [Setting Up Oracle Cloud Database-5.docx](#)[Download Setting_Up_Oracle_Cloud_Database-5.docx](#)

[Setting_Up_Oracle_Cloud_Database-5.docx](#)

Notice that it is advised that you create a new user, so not use the ADIN user for your database tables.

OPTIONAL: if you wish to connect your OCI to Oracle SQL Developer you can use the instructions here: [Connect OCI with developer-2.rtf](#)[Download Connect_OCI_with_developer-2.rtf](#)

[Connect_OCI_with_developer-2.rtf](#)

Step 2

Use your solution to the previous lab. Open the relational schema, then Ctrl+A to select all the relations and right click to DDL Preview. If you have errors in the scripts, you may need to go back to the ER diagram and

define datatypes, PKs and other details. Once you are ready and no longer have errors, copy all the DDL scripts and paste them in your OCI database worksheet window.

Step 3

Execute all scripts to create the relations. Have you created tables with PKs and FKs?

The screenshot displays the Oracle Cloud Database Actions interface. The left sidebar shows a tree view of database objects under the 'KOREY' schema, including tables like FEATURE_ID, FEATURE, CAR_VIN, PAYMENT, and RENTAL. The main area shows SQL scripts for creating and altering tables. The 'Query Result' tab is active, showing a table with columns RENTAL_ID, START_DATE, RETURN_DATE, TOTAL_COST, CUSTOMER_CUST, and CAR_VIN. The table contains one row of data.

```
1 CREATE TABLE address (
2   address_type CHAR(10) NOT NULL,
3   street VARCHAR2(20),
4   "Number" NUMBER(11),
5   city VARCHAR2(20),
6   state CHAR(2),
7   zip VARCHAR2(10),
8   customer_customer_id INTEGER NOT NULL
9 );
10
11 ALTER TABLE address ADD CONSTRAINT address_pk PRIMARY KEY ( address_type,
12   street, "Number", city, state, zip, customer_customer_id );
13
14 CREATE TABLE car (
15   vin INTEGER NOT NULL,
16   make VARCHAR2(20),
17   model VARCHAR2(20) NOT NULL,
18   car_type_car_type_id INTEGER NOT NULL
19 );
20
21 ALTER TABLE car ADD CONSTRAINT car_pk PRIMARY KEY ( vin );
```

RENTAL_ID	START_DATE	RETURN_DATE	TOTAL_COST	CUSTOMER_CUST	CAR_VIN
1	12/25/2025, 12:0...	12/27/2025, 12:0...	600	1	123456

Execution time: 0.008 seconds

3 3 0 0 9:30:14 PM - 1 rows total

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Step 4

Insert at least 1 record in each table. Do the data make sense? Try to insert data into CarFeatures before you insert data into Car. Can you do that? Explain. Don't forget to commit your changes.

The data does make sense. When I tried to insert data into carFeatures before I inserted data into car, I was not able to do that. The error I got indicated that the carFeatures has a composite primary key and the car it is trying to reference does not exist yet so it is not possible to add the data to that key.

If you are trying to insert dates, you need to transform the character input to a date, so the command will be like:

```
INSERT INTO ... VALUES ( ..., TO_DATE('01/02/2015', 'MM/DD/YYYY'), ... )
```

The TO_DATE function transforms the input character: '01/02/2015' to a date with MM=01, DD=02 and YYYY=2015

The screenshot shows the Oracle Database Actions interface. The left sidebar displays a tree view of the database schema, including tables like FEATURE_ID, FEATURE, CAR_VIN, PAYMENT, PREFERENCE, and RENTAL. The main area shows the SQL script being executed, which includes INSERT statements for CUSTOMER, ADDRESS, CAR_TYPE, CAR, RENTAL, PREFERENCE, and PAYMENT, followed by a SELECT statement to verify the data. The query results are displayed in a table below the script.

RENTAL_ID	START_DATE	RETURN_DATE	TOTAL_COST	CUSTOMER_CUST	CAR_VIN
1	12/25/2025, 12:00...	12/27/2025, 12:00...	600	1	123456

Execution time: 0.008 seconds

Step 5

Now let's try some basic DML commands:

1. Update one of the car records you have added and change one of the values.
2. Delete a record of your choice.
3. Write an update statement that will increase all payment amounts prices by 10%

Don't forget to commit your changes.

The screenshot shows the Oracle Database Actions interface. The left sidebar displays a tree view of database objects under the 'KOREY' schema, including tables like CAR_TYPE, CUSTOMER, FEATURE, PAYMENT, and PREFERENCE. The main area shows a SQL script with the following commands:

```
1 UPDATE CAR SET MAKE = 'Honda', MODEL = 'HRV' WHERE CAR_TYPE_CAR_TYPE_ID = 1;
2
3 -- Adding data to delete
4 INSERT INTO CAR_TYPE VALUES(2, 'This is a truck');
5 INSERT INTO CAR VALUES(98765, 'Ford', 'Tacoma', 2);
6
7 -- Delete Data
8 DELETE FROM CAR WHERE VIN = 98765;
9
10
11
12 -- Update Payments
13 UPDATE PAYMENT SET PAY_AMOUNT = PAY_AMOUNT * 1.10;
14
15
16 SELECT * FROM PAYMENT;
17 SELECT * FROM CAR;
18
19 commit;
```

Below the script, the 'Query Result' tab shows the execution time as 0.003 seconds. A table displays the results of the SELECT queries:

PAYMENT_ID	PAY_DATE	PAY_TIME	PAY_AMOUNT	RENTAL_RENTAL_
1	12/27/2025, 12:0...	2026-02-01T00:...	660	1

Step 6

Now let's try some basic DDL commands:

1. Add a new column of your choice to one of your tables.
2. Change the datatype of a column of your choice
3. Modify a table of your choice and change one of the column definitions to a column that cannot be NULL
4. Modify a table of your choice and change one of the column definitions to a column has a default value 0
5. Truncate one of your tables. Can you still see the table definition? What about the data?

After truncating my feature table I was unable to see my table definition, and I could also not see any of my data that I added to the table.

6. Drop one of your tables. Can you still see the table definition? Can you drop any of your tables? Try dropping Cars table. What happens and why?

I chose to drop the Feature table and after doing so successfully I was no longer able to view the table definition anymore. Yes, I was able to drop a table and that was the feature table like I previously mentioned this is because there is no f keys or anything

that relies on it and I am able to drop it. When I try and drop the cars table I get an error this is because there are tables in my schema that reference columns in car such as rental which uses the vin from car.

The screenshot displays the Oracle SQL Developer interface. On the left, the 'Connections' pane shows a connection to 'CIS2109'. The 'Tables (Filtered)' list includes ADDRESS, CAR, CAR_TYPE, CUSTOMER, DBTOOLS\$EXECUTION_HISTORY, PAYMENT, PREFERENCE, RENTAL, Views, Indexes, Packages, Procedures, Functions, Operators, Queues, and Queues Tables. The 'Reports' pane lists various report types like All Reports, About Your Database, All Objects, etc.

The main window shows a 'Worksheet' with a SQL script. The script includes comments and SQL commands for altering tables, updating data, truncating a table, and dropping tables. The 'Script Output' pane shows the execution results, indicating that the task was completed in 0.11 seconds. A message states: 'Before performing the above operations the table, drop the foreign key constraints in other tables. You can see what constraints are referencing a table by issuing the following command: SELECT * FROM USER_CONSTRAINTS WHERE TABLE_NAME = "tabnam";'. The output also shows 'Commit complete.' and a table of constraints.

Name	Null?	Type
VIN	NOT NULL	NUMBER(38)
MAKE		VARCHAR2(20)
MODEL	NOT NULL	VARCHAR2(20)
CAR_TYPE_CAR_TYPE_ID	NOT NULL	NUMBER(38)
PRICE		NUMBER(10,2)